

For each of the following statement, answer whether the statement is true or false.

(a) $1000n + n \log n = O(n \log n)$.

(b) $n^2 + n \log(n^3) = O(n \log(n^3))$.

(c) $n^3 = \Omega(n)$.

(d) $n^2 + n = \Omega(n^3)$.

(e) $n^3 = O(n^{10})$.

(f) $n^3 + 1000n^{2.9} = \Theta(n^3)$

(g) $n^3 - n^2 = \Theta(n)$

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- | | |
|--|------------|
| (a) $1000n + n \log n = O(n \log n)$. | (a) True. |
| (b) $n^2 + n \log(n^3) = O(n \log(n^3))$. | (b) False. |
| (c) $n^3 = \Omega(n)$. | (c) True. |
| (d) $n^2 + n = \Omega(n^3)$. | (d) False. |
| (e) $n^3 = O(n^{10})$. | (e) True. |
| (f) $n^3 + 1000n^{2.9} = \Theta(n^3)$ | (f) True. |
| (g) $n^3 - n^2 = \Theta(n)$ | (g) False. |

For each pair of expressions (A, B) below, indicate whether A is O , Ω , or Θ of B . Note that zero, one, or more of these relations may hold for a given pair; list all correct ones. Justify your answers.

(a) $A = n^3 + n \log n$; $B = n^3 + n^2 \log n$.

(b) $A = \log \sqrt{n}$; $B = \sqrt{\log n}$.

(c) $A = n \log_3 n$; $B = n \log_4 n$.

(d) $A = 2^n$; $B = 2^{n/2}$.

(e) $A = \log(2^n)$; $B = \log(3^n)$.

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	A	Relation:	B
(a)	$n^3 + n \log n$	Ω, Θ, O	$n^3 + n^2 \log n$
(b)	$\log \sqrt{n}$	Ω	$\sqrt{\log n}$
(c)	$n \log_3 n$	Ω, Θ, O	$n \log_4 n$
(d)	2^n	Ω	$2^{n/2}$
(e)	$\log(2^n)$	Ω, Θ, O	$\log(3^n)$

Notes:

- (a) Both are $\Theta(n^3)$, the lower order terms can be ignored.
Note that if $A(n) = \Theta(B(n))$, then automatically $A(n) = O(B(n))$ and $A(n) = \Omega(B(n))$.
- (b) After simplifying, A is $(1/2) \log n$, and B is $\sqrt{\log n}$.
Substituting $m = \log n$, we can see ratio A/B grows as $m/2\sqrt{m} = \sqrt{m}/2$ which tends to infinity as n (and hence m) tends to infinity, i.e., $A(n) = \Omega(B(n))$.
- (c) Log base conversion only introduces a constant factor.
- (d) The ratio is $2^n/2^{n/2} = (2)^{n/2}$ which goes to infinity in the limit.
- (e) After simplifying these are $n \log 2$ and $n \log 3$, both of which are $\Theta(n)$.

Suppose $T_1(n) = O(f(n))$ and $T_2(n) = O(f(n))$. Which of the following are true? Justify your answers.

(a) $T_1(n) + T_2(n) = O(f(n))$

(b) $\frac{T_1(n)}{T_2(n)} = O(1)$

(c) $T_1(n) = O(T_2(n))$

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(a) True. Since $T_1(n) = O(f(n))$ and $T_2(n) = O(f(n))$, it follows from the definition that there exist constants $c_1, c_2 > 0$ and positive integers n_1, n_2 such that $T_1(n) \leq c_1 f(n)$ for $n \geq n_1$ and $T_2(n) \leq c_2 f(n)$ for $n \geq n_2$. This implies that, $T_1(n) + T_2(n) \leq (c_1 + c_2)f(n)$ for $n \geq \max(n_1, n_2)$. Thus, $T_1(n) + T_2(n) = O(f(n))$.

(b) False. For a counterexample to the claim, let $T_1(n) = n^2$, $T_2(n) = n$, $f(n) = n^2$. Then $T_1(n) = O(f(n))$ and $T_2(n) = O(f(n))$ but $\frac{T_1(n)}{T_2(n)} = n \neq O(1)$.

(c) False. We can use the same counterexample as in part (b). Note that $T_1(n) \neq O(T_2(n))$

Let $f(n)$ and $g(n)$ be non-negative functions. Using the basic definition of Θ -notation, prove that $\max(f(n), g(n)) = \Theta(f(n) + g(n))$.

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For any value of n , $\max(f(n), g(n))$ is either equal to $f(n)$ or equal to $g(n)$. Therefore, for all n ,

$$\max(f(n), g(n)) \leq f(n) + g(n).$$

Using $c = 1$ and $n_0 = 1$ in the big-oh definition, it follows that

$$\max(f(n), g(n)) = O(f(n) + g(n)).$$

Also, for all n ,

$$\max(f(n), g(n)) \geq f(n)$$

and

$$\max(f(n), g(n)) \geq g(n).$$

Adding we have

$$2 \times \max(f(n), g(n)) \geq f(n) + g(n).$$

Therefore,

$$\max(f(n), g(n)) \geq \frac{1}{2}(f(n) + g(n)).$$

Using $c = 1/2$ and $n_0 = 1$ in the Omega definition, it follows that

$$\max(f(n), g(n)) = \Omega(f(n) + g(n)).$$

Since $\max(f(n), g(n)) = O(f(n) + g(n))$ and $\max(f(n), g(n)) = \Omega(f(n) + g(n))$, it implies that $\max(f(n), g(n)) = \Theta(f(n) + g(n))$.